CLI Usage

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1. CLI Usage Syntax

CLI (Command-Line Interface) is a way for users to interact with computers or software programs. Users run programs or perform tasks by typing text commands instead of clicking icons or buttons through a graphical user interface (GUI).

```
yolo TASK MODE ARGS

Where   TASK (optional) is one of [detect, segment, classify, pose, obb]
        MODE (required) is one of [train, val, predict, export, track,
benchmark]
        ARGS (optional) are any number of custom 'arg=value' pairs like
'imgsz=320' that override defaults.
```

2. Start

2.1. Enter docker

Run YOLOv11's docker script

```
sh ~/yolov11_dcoker.sh
```

2.2. Image prediction

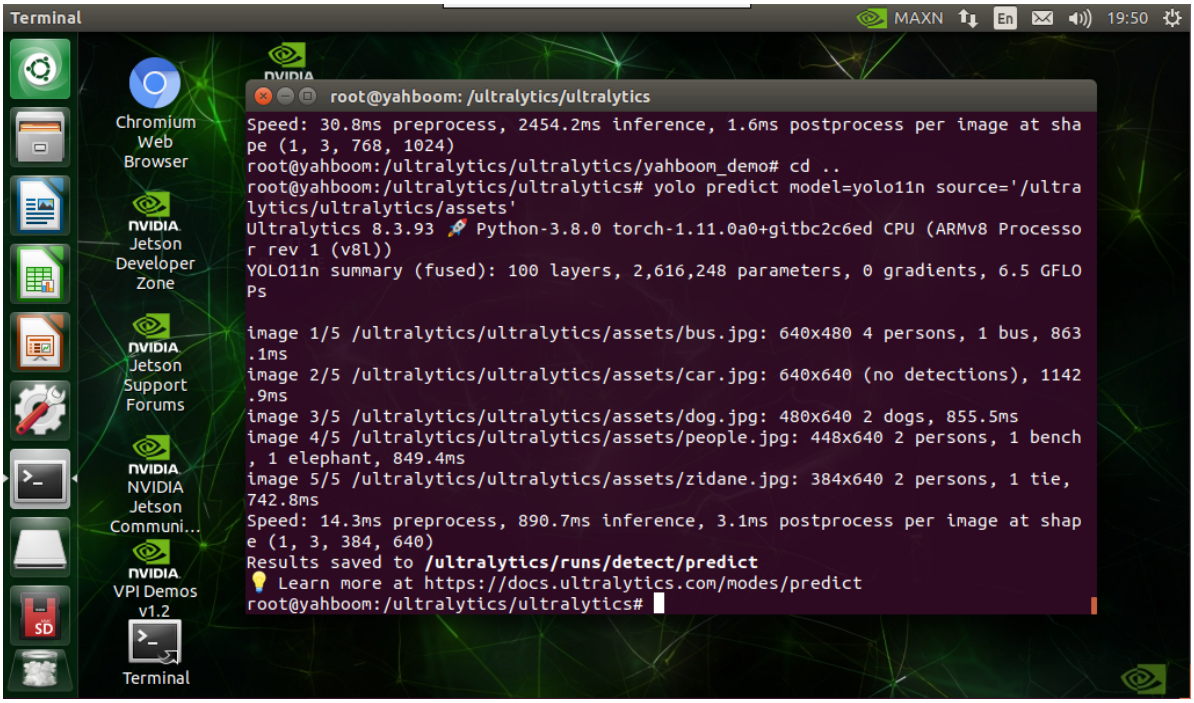
Use yolo11n.pt to predict the images provided by the ultralytics project: If the system does not find the corresponding model file in the directory where the command is run, it will automatically download it (if it cannot be downloaded, you can copy the model into it yourself)

Enter the project folder:

```
cd /ultralytics/ultralytics/
```

Use yolo11n.pt to detect the images in the target folder and output the results:

```
yolo predict model=yolo11n source='/ultralytics/ultralytics/assets'
```



Effect preview

Video location of yolo recognition output: /ultralytics/runs/detect;

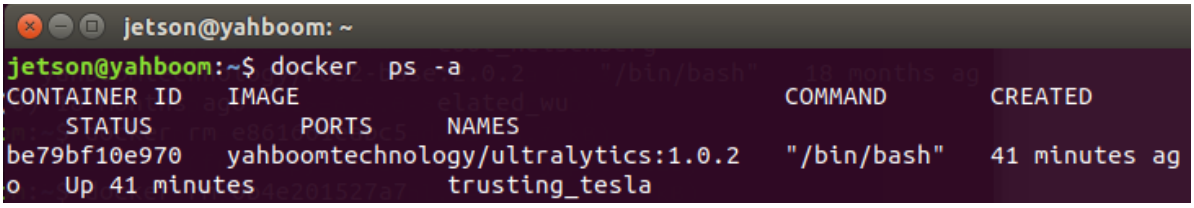
Each time it runs, the predict in this folder automatically increases by 1

There are two preview methods:

1. View using jupyter lab

Open another terminal to enter the docker container and use jupyter lab to view the image

```
docker ps -a
```

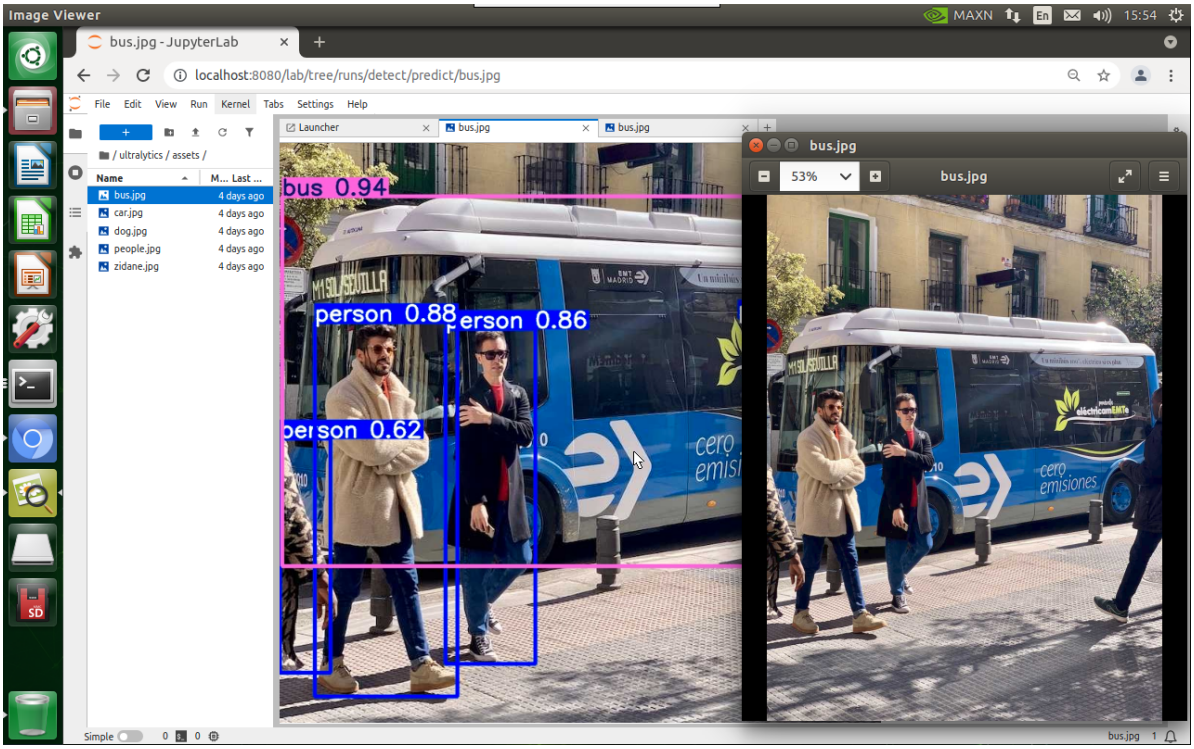



```
docker exec -it be79bf10e970 /bin/bash #Container ID needs to be modified according to your actual search
cd /ultralytics
jupyter lab --allow-root
```

```
root@yahboom: /ultralytics
root@yahboom:/ultralytics# jupyter lab --allow-root
[I 2025-03-24 07:38:52.970 ServerApp] jupyter_lsp | extension was successfully l
inked.
[I 2025-03-24 07:38:52.985 ServerApp] jupyter_server_terminals | extension was s
uccessfully linked.
[I 2025-03-24 07:38:53.011 ServerApp] jupyterlab | extension was successfully li
nked.
[I 2025-03-24 07:38:53.776 ServerApp] notebook_shim | extension was successfully
 linked.
[I 2025-03-24 07:38:53.865 ServerApp] notebook_shim | extension was successfully
 loaded.
[I 2025-03-24 07:38:53.875 ServerApp] jupyter_lsp | extension was successfully l
oaded.
[I 2025-03-24 07:38:53.879 ServerApp] jupyter_server_terminals | extension was s
uccessfully loaded.
[I 2025-03-24 07:38:53.883 LabApp] JupyterLab extension loaded from /usr/local/l
ib/python3.8/dist-packages/jupyterlab
[I 2025-03-24 07:38:53.884 LabApp] JupyterLab application directory is /usr/loca
l/share/jupyter/lab
[I 2025-03-24 07:38:53.885 LabApp] Extension Manager is 'pypi'.
[I 2025-03-24 07:38:54.000 ServerApp] jupyterlab | extension was successfully lo
aded.
[I 2025-03-24 07:38:54.001 ServerApp] Serving notebooks from local directory: /u
ltralytics
```

Access directly through http://localhost:8080/ in the system browser:

```
http://localhost:8080/
```



2. Copy the file to the host machine for viewing

Enter the following command in the host terminal

```
docker cp be79bf10e970: /ultralytics/runs/detect /home/jetson/ultralytics/runs
#Container ID needs to be modified according to the actual one you find
```

```
jetson@yahboom:~$ docker cp be79bf10e970:/ultralytics/runs/detect /home/jetson/u
ltralytics/runs
```



2.3, Video Prediction

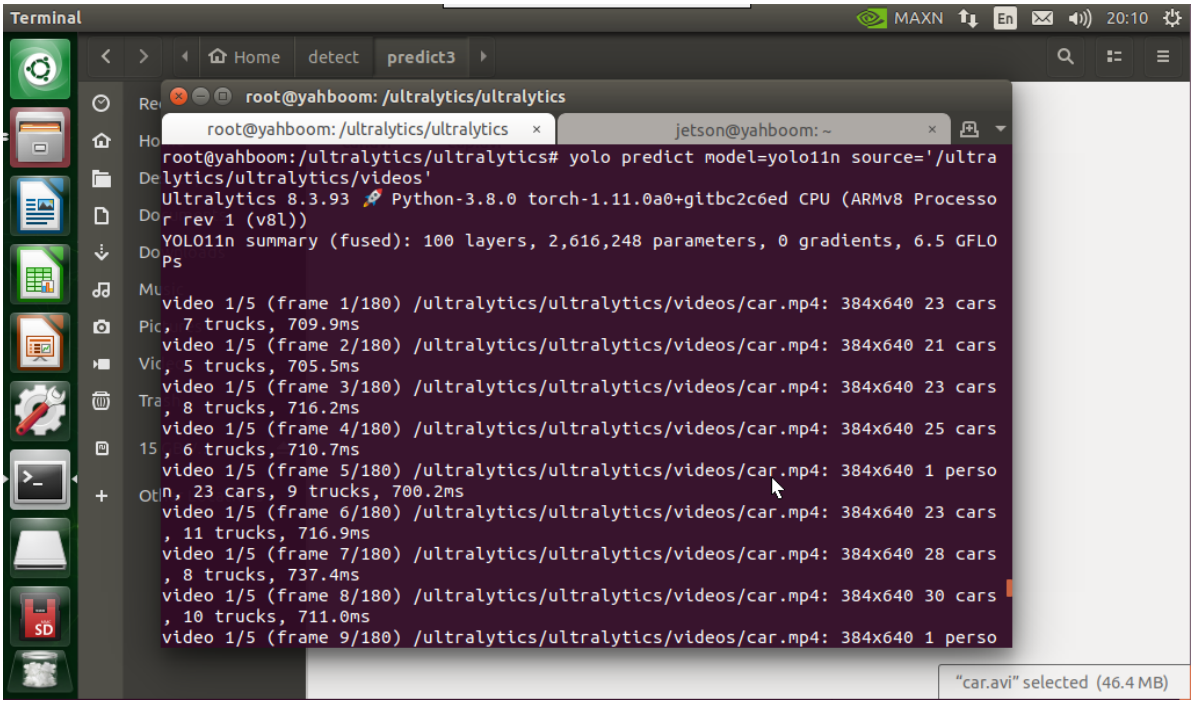
Use yolo11n.pt to predict videos under the ultralytics project (not ultralytics built-in videos): If the system does not find the corresponding model file in the directory where the command is run, it will automatically download it (if it cannot be downloaded, you can copy the model into it yourself)

Enter the project folder:

```
cd /ultralytics/ultralytics
```

Use yolo11n.pt to detect the video in the target folder and output the result:

```
yolo predict model=yolo11n source='/ultralytics/ultralytics/videos'
```



Effect preview

Video location of yolo recognition output: /ultralytics/runs/detect

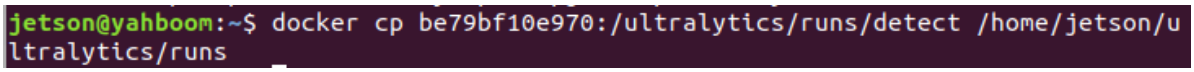
Each time it runs, the predict in this folder automatically increases by 1

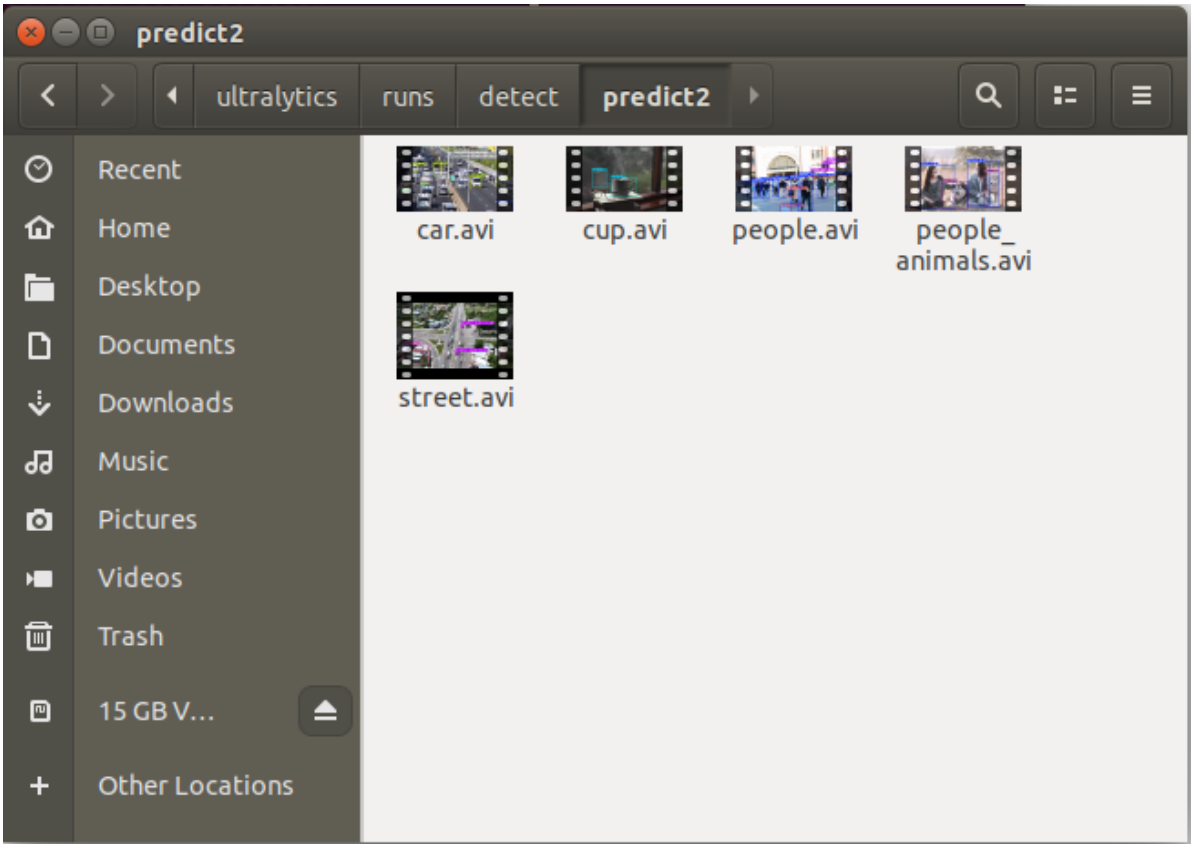
The output video format is .avi file, which cannot be viewed in jupyter lab. You can copy the file to the host machine for viewing.

For example:

Enter the following command in the host terminal

```
docker cp be79bf10e970:/ultralytics/runs/detect /home/jetson/ultralytics/runs
#Container ID needs to be modified according to the actual one you find
```







2.4, Real-time prediction

Use yolo11n.pt to predict the USB camera screen: If the system does not find the corresponding model file in the directory where the command is run, it will automatically download it (if it cannot be downloaded, you can copy the model yourself)

Enter the project folder:

```
cd /ultralytics/ultralytics
```

Use yolo11n.pt to detect the camera screen and output the results: Currently, only USB cameras can directly use CLI to predict real-time images, and CSI cameras do not have relevant information that can be used directly as input sources

```
yolo predict model=yolo11n.pt source=0 save=False show # Object detection
# Instance segmentation: yolo predict model=yolo11n-seg.pt source=0 save=False
show
# Image classification: yolo predict model=yolo11n-cls.pt source=0 save=False
show
# Pose estimation: yolo predict model=yolo11n-pose.pt source=0 save=False show
# Oriented bounding box object detection: yolo predict model=yolo11n-obb.pt
source=0 save=False show
```

Click the terminal and press the "Ctrl + C" shortcut key to terminate the program!

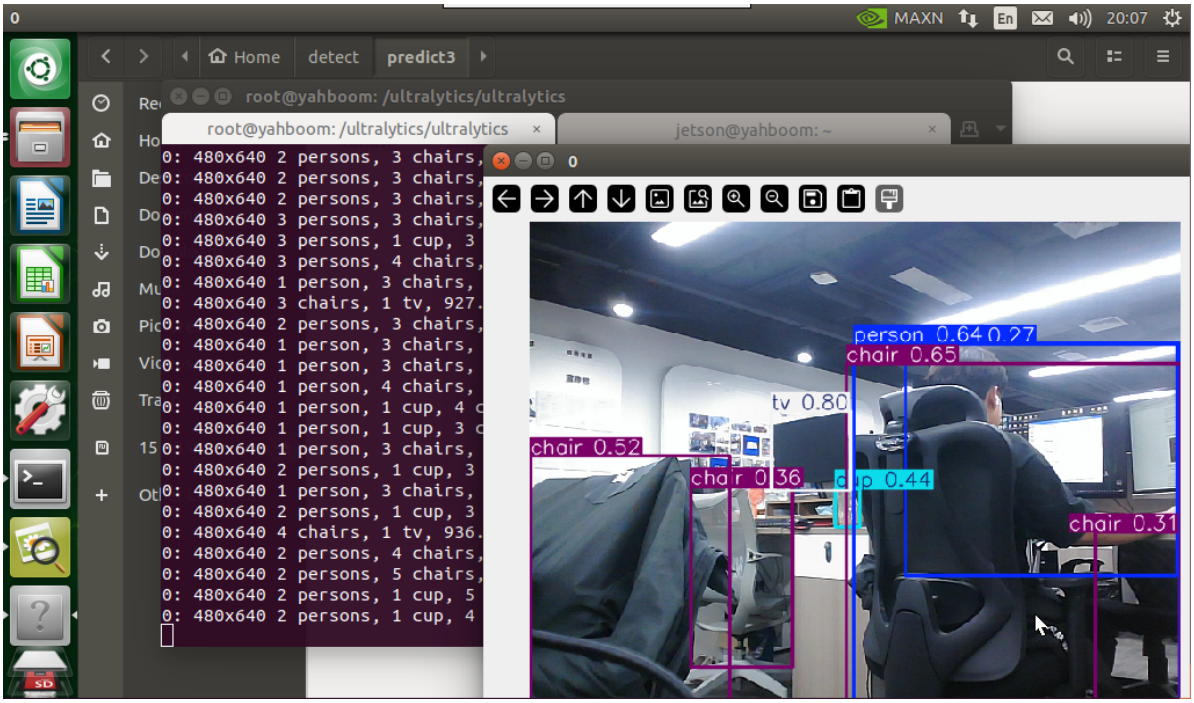
> **Parameter Description**

`model`: Specify the YOLO model

`source`: Specify the recognition source: multiple cameras can switch numbers

`save=False`: Disable saving results

`show`: Real-time display



References

https://github.com/ultralytics/ultralytics

https://docs.ultralytics.com/usage/cli/